Department of Mathematics
Motilal Nehru National Institute of Technology Allahabad
Mathematics II (MAN12101)
Course Instructor: Dr. Naren Bag
Assignment 5(Fourier Series and Fourier Transform)

1. Write the sufficient conditions for the existence of Fourier series for a function $f(x)$. Also, obtain the Fourier series of the function $f(x) = \begin{cases} 1+2x/\pi, & -\pi \le x \le 0 \\ 1-2x/\pi, & 0 \le x \le \pi \end{cases}$ and deduce that $1 + \frac{1}{3^2} + + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + - - - - - - - - - = \frac{\pi^2}{8}$.

2. Find the Fourier series expansion of the function $f(x) = x^2,\ -2 \le x \le 2$. Hence prove that

$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ and $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2} = \frac{\pi^2}{12}$.

3. Find the Fourier series expansion of $f(x) = \begin{cases} \pi + x, & -\pi \le x \le 0 \\ 0, & 0 \le x \le \pi \end{cases}$, $f(x+2\pi) = f(x)$.

4. Find the Fourier series expansion for the periodic function $f(x)$ defined by $f(x) = \begin{cases} 1-x, & -\pi \le x \le 0 \\ 1+x, & 0 \le x \le \pi \end{cases}$.
Also, deduce that $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$.

5. Find the Fourier series to represent $f(x) = \frac{\pi - x}{2}$ in the interval $(0, 2\pi)$ and hence deduce $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + - - - - - -$

6. Find the Fourier series of period 2π for the function $f(x) = x(2\pi - x)$ in $(0, 2\pi)$. Deduce the sum of the series $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + ...\infty$.

7. Find the Fourier series of period $2l$ for the function $f(x) = |\, x\, |$ in $(-l, l)$. Hence find the value of $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + ...\infty$.

8. Find the Fourier series of period 2π for the function $f(x) = |\sin x|$ in $(-\pi, \pi)$.

9. Find the Fourier series of period 2π for the function $f(x) = \cos ax$ in $(-\pi, \pi)$, where a is not an integer. Deduce that the sum of the series $\sum_{n=1}^{\infty} \frac{1}{9n^2-1}$.

10. Obtain half range cosine series for $f(x) = \begin{cases} kx, & 0 \le x \le l/2 \\ k(l-x), & l/2 \le x \le l \end{cases}$. Deduce the sum of the series $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + ...$

11. Find the half-range sine series of $f(x) = x \sin x$ in $(0, \pi)$.

12. Find the half-range cosine series of $f(x) = 6x^2 - 6x + 1$ in $0 < x < 1$. Deduce the sum of the series $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + ...\infty$.

13. Find the half-range sine series of $f(x) = x \cos \pi x$ in $0 < x < 1$. Deduce the sum of the series $\frac{1}{1.2} - \frac{1}{2.3} + \frac{1}{3.4} - ....\infty$.

14. Find the half-range sine series of $f(x) = \pi/2 - x$ in $(0, \pi)$. Deduce the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

15. Find the Fourier cosine integral representation of $f(x) = e^{-kx},\ x \ge 0$, where k is a positive constant.

16. Find the Fourier cosine integral representation of $f(x) = \begin{cases} \sin x, & 0 \le x \le \pi \\ 0, & x > \pi \end{cases}$.

17. Find the Fourier cosine and sine integral of the function $f(x) = e^{-\lambda x},\ x \ge 0$, where λ is a positive constant. Hence deduce $\int_0^{\infty} \frac{\cos sx}{1+s^2} ds$.

18. Find the Fourier integral representation of the piecewise continuous function $f(x) = \begin{cases} 0, & x < 0 \\ 1, & 0 < x < 2 \\ 0, & x > 2 \end{cases}$

19. Find the Fourier integral representation of the function $f(x) = \begin{cases} e^x, & |x| < 2 \\ 0, & |x| > 2 \end{cases}$.

20. Find the complex form of fourier integral representation for the function $f(x) = \begin{cases} |x|, & -\pi < x < \pi \\ 0, & \text{otherwise} \end{cases}$.

21. Find the complex form of fourier integral representation for the function $f(x) = \begin{cases} \sinh x, & |x| < a \\ 0, & |x| > a \end{cases}$.

22. Find the Fourier transform of $f(x) = \begin{cases} 1, & |x| \leq a \\ 0, & |x| > a \end{cases}$. Hence evaluate the value of the integral $\int_{-\infty}^{\infty} \frac{\sin(pa)\cos(px)}{p} dp$ and $\int_0^{\infty} \frac{\sin p}{p} dp$.

23. Find the Fourier transform of $\frac{\sin ax}{x}$ and hence find the value of $\int_{-\infty}^{\infty} \frac{\sin^2 ax}{x^2} dx$.

24. Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$. Hence evaluate $\int_0^{\infty} \left(\frac{\sin x - x\cos x}{x^3}\right) \cos\left(\frac{x}{2}\right) dx$.

25. Find the Fourier sine transform of $f(x) = e^{-ax}$ for $x \geq 0$ and $a \geq 0$. Deduce the integral

$\int_0^{\infty} \frac{p\sin(px)}{a^2+p^2} dp$ and show that $\int_0^{\infty} \frac{\sin(px)}{p} dp = \frac{\pi}{2}$. Also, find $F_s\{xe^{-ax}\}$ and $F_s\{\frac{e^{-ax}}{x}\}$.

26. Find the inverse Fourier transform of

i) $\frac{s}{s^2+1}$ ii) $\frac{1}{(s^2+1)^2}$

27. Find the inverse Fourier transform of $F(s) = \begin{cases} a - |s|, & \text{if } |s| \leq a \\ 0, & |s| > a \end{cases}$. Hence evaluate $\int_0^{\infty} \frac{\sin^2 x}{x^2} dx$.

28. Find the inverse Fourier cosine transform of e^{-x}

29. Find the finite Fourier sine transform of $\cos ax$ and finite Fourier cosine transform of $\sin ax$ in $(0, \pi)$.

30. Using convolution find the inverse of the following functions

i) $\frac{1}{(is+a)^2},\ a > 0$ ii) $\frac{1}{(is+a)^3},\ a > 0$

31. Using Parseval's identity, show that

i) $\int_0^{\infty} \frac{t^2 dt}{(4+t^2)(9+t^2)} = \frac{\pi}{10}$ ii) $\int_0^{\infty} \frac{\sin(at)dt}{t(a^2+t^2)} = \frac{\pi}{2}\frac{(1-e^{-a^2})}{a^2}$

32. Find the Fourier transform of $f(x)$, given by $f(x) = \begin{cases} a - |x|, & |x| < a \\ 0, & |x| > a \end{cases}$.

Hence show that $\int_0^{\infty} \left(\frac{\sin t}{t}\right)^2 dt = \frac{\pi}{2}$ and $\int_0^{\infty} \left(\frac{\sin t}{t}\right)^4 dt = \frac{\pi}{3}$.

33. Find the solution of the following differential equations

i) $y' - 4y = H(t)e^{-4t},\ -\infty < t < \infty$

ii) $y'' + 3y' + 2y = \delta(t-3)$.

34. Find the solution of the following initial boundary value problems

i) $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2},\ 0 < x < \infty,\ t > 0$

$u(x,0) = \begin{cases} 1, & 0 < x \leq l \\ 0, & x > l \end{cases},\quad u(0,t) = 0,\ t > 0.$

ii) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,\ -\infty < x < \infty,\ 0 < y < \pi$

$u(x,0) = e^{-2x}H(x),\quad u(x,\pi) = 0, -\infty < x < \infty.$